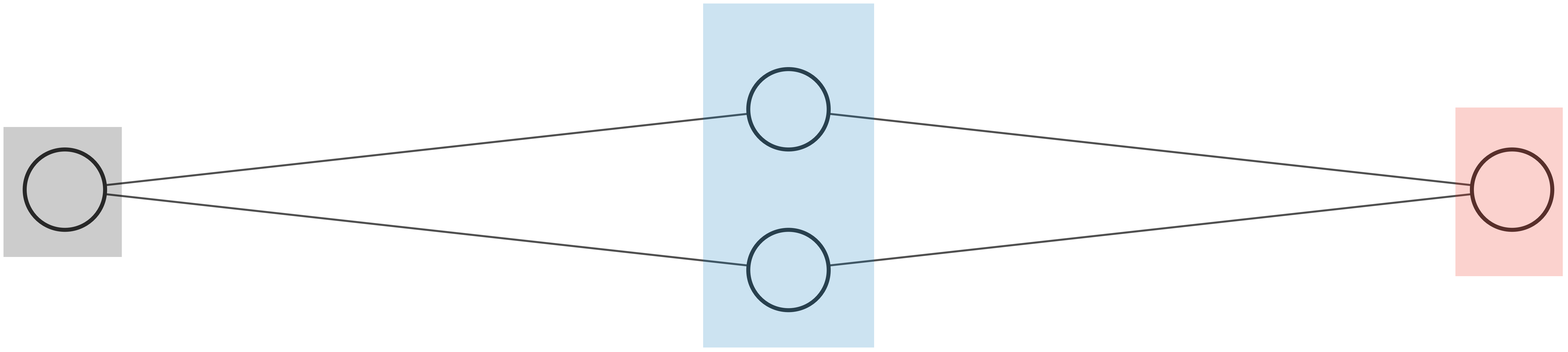
Let’s see some examples of
How you can construct more complex
Neural Network with Julia using the Lux()
package

```
...    Chain(
           layer_1 = Dense(1 => 2, tanh),              # 4 parameters
           layer_2 = Dense(2 => 1),                    # 3 parameters
       )         # Total: 7 parameters,
                 #        plus 0 states.
```

```
model = Chain(Dense(1 => 2,tanh),
              Dense(2=>1))
```



Input Layer $\in \mathbb{R}^1$

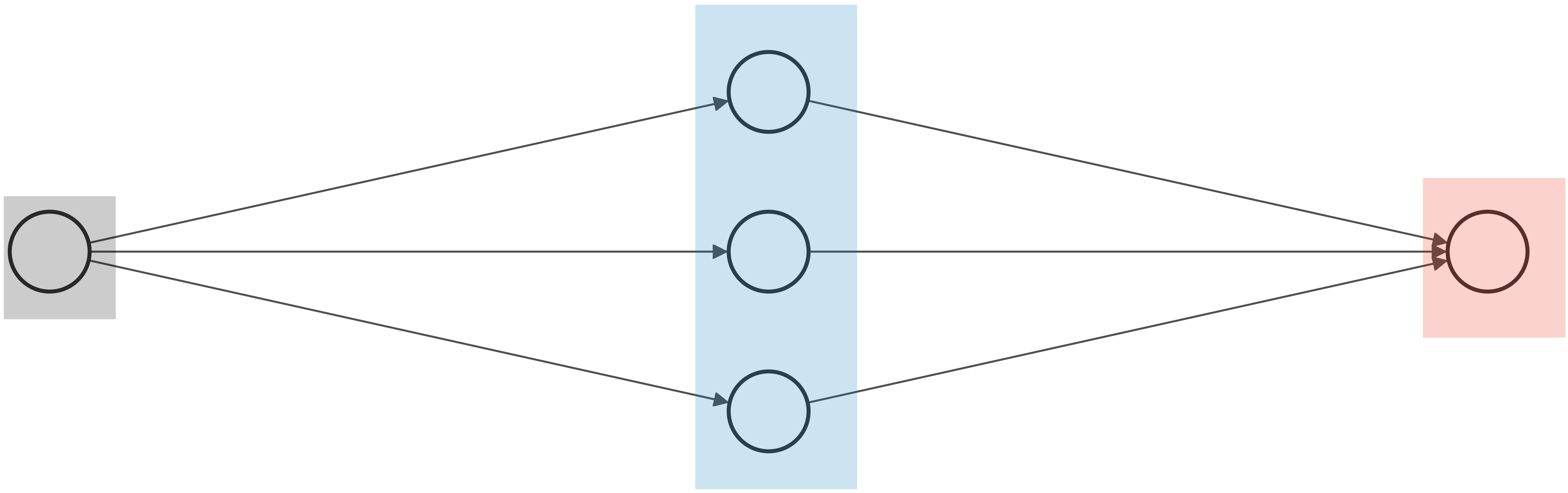
Hidden Layer $\in \mathbb{R}^2$

Output Layer $\in \mathbb{R}^1$

$$y(x) = NN(x) = W_{21} \tanh(W_{11}x + b_{11}) + W_{22} \tanh(W_{12}x + b_{12}) + b_{21}$$

```
Chain(
    layer_1 = Dense(1 => 3, tanh),                # 6 parameters
    layer_2 = Dense(3 => 1),                      # 4 parameters
)         # Total: 10 parameters,
          #        plus 0 states.
```

```
model = Chain(Dense(1 => 3,tanh),
              Dense(3=>1))
```



Input Layer $\in \mathbb{R}^1$

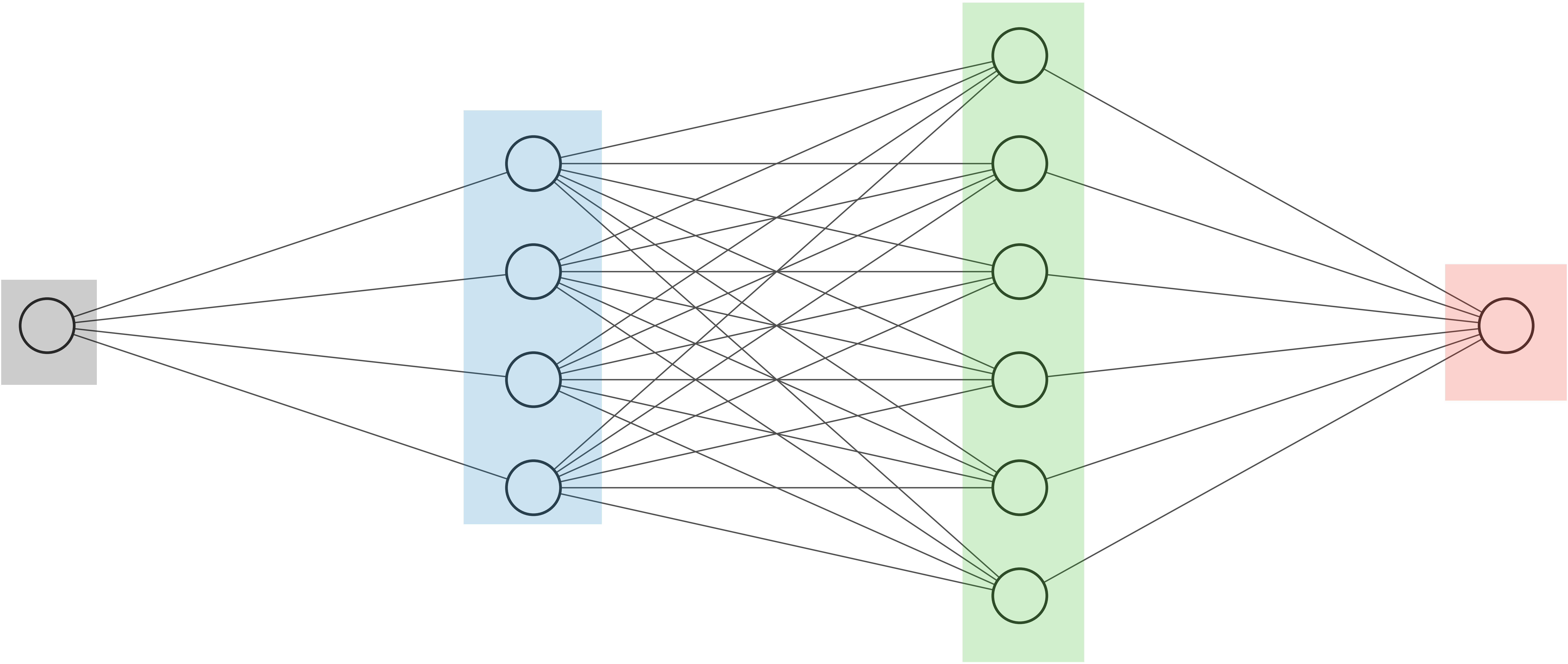
Hidden Layer $\in \mathbb{R}^3$

Output Layer $\in \mathbb{R}^1$

$$W_{21}\tanh(W_{11}x + b_{11}) + W_{22}\tanh(W_{12}x + b_{12}) + W_{23}\tanh(W_{13}x + b_{13}) + b_{21}$$

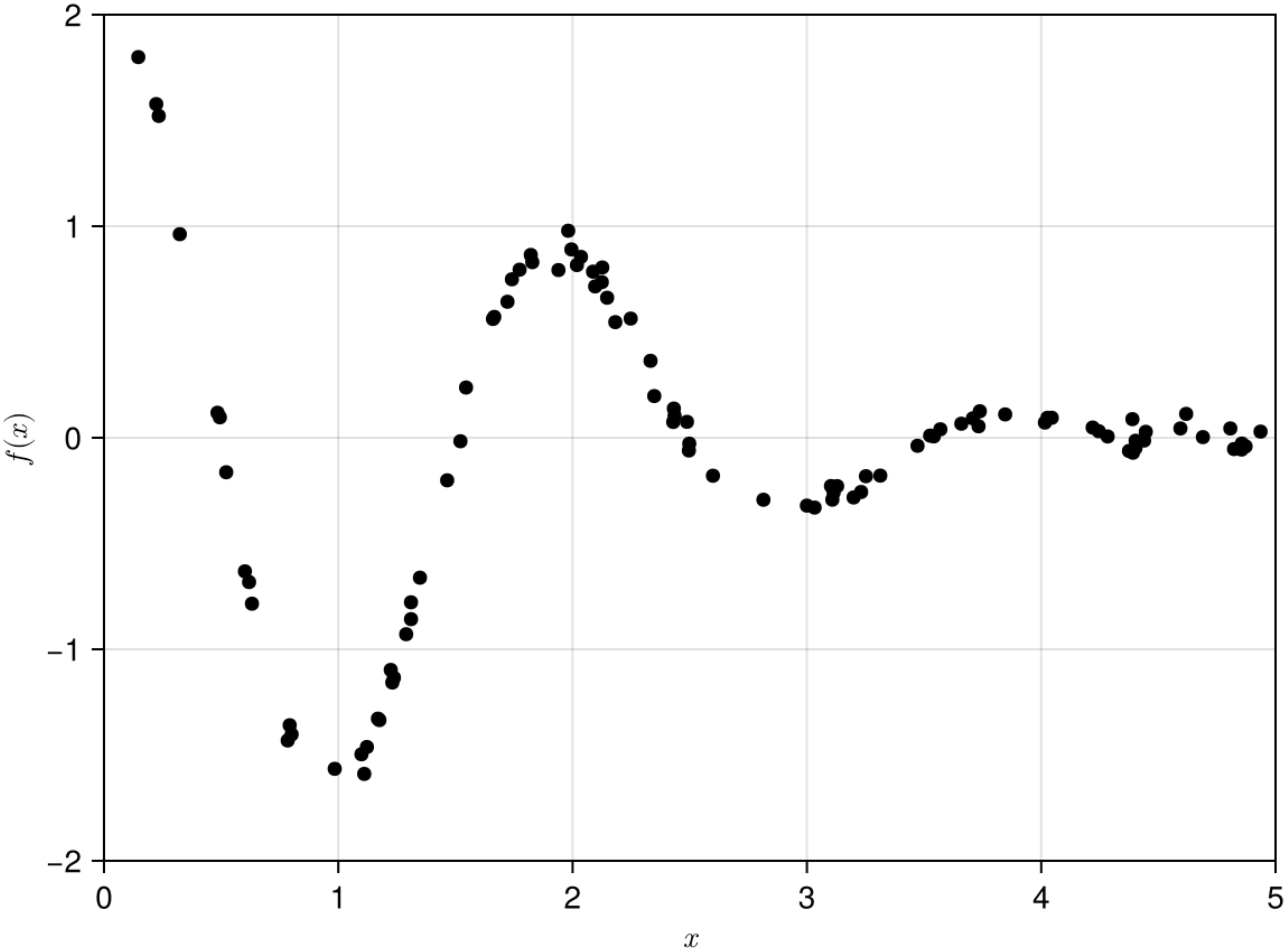
```
Chain(
    layer_1 = Dense(1 => 4, tanh),                # 8 parameters
    layer_2 = Dense(4 => 6, tanh),                # 30 parameters
    layer_3 = Dense(6 => 1),                      # 7 parameters
)         # Total: 45 parameters,
          #        plus 0 states.
```

```
model = Chain(Dense(1 => 4,tanh),
              Dense(4=>6,tanh),
              Dense(6=>1))
```

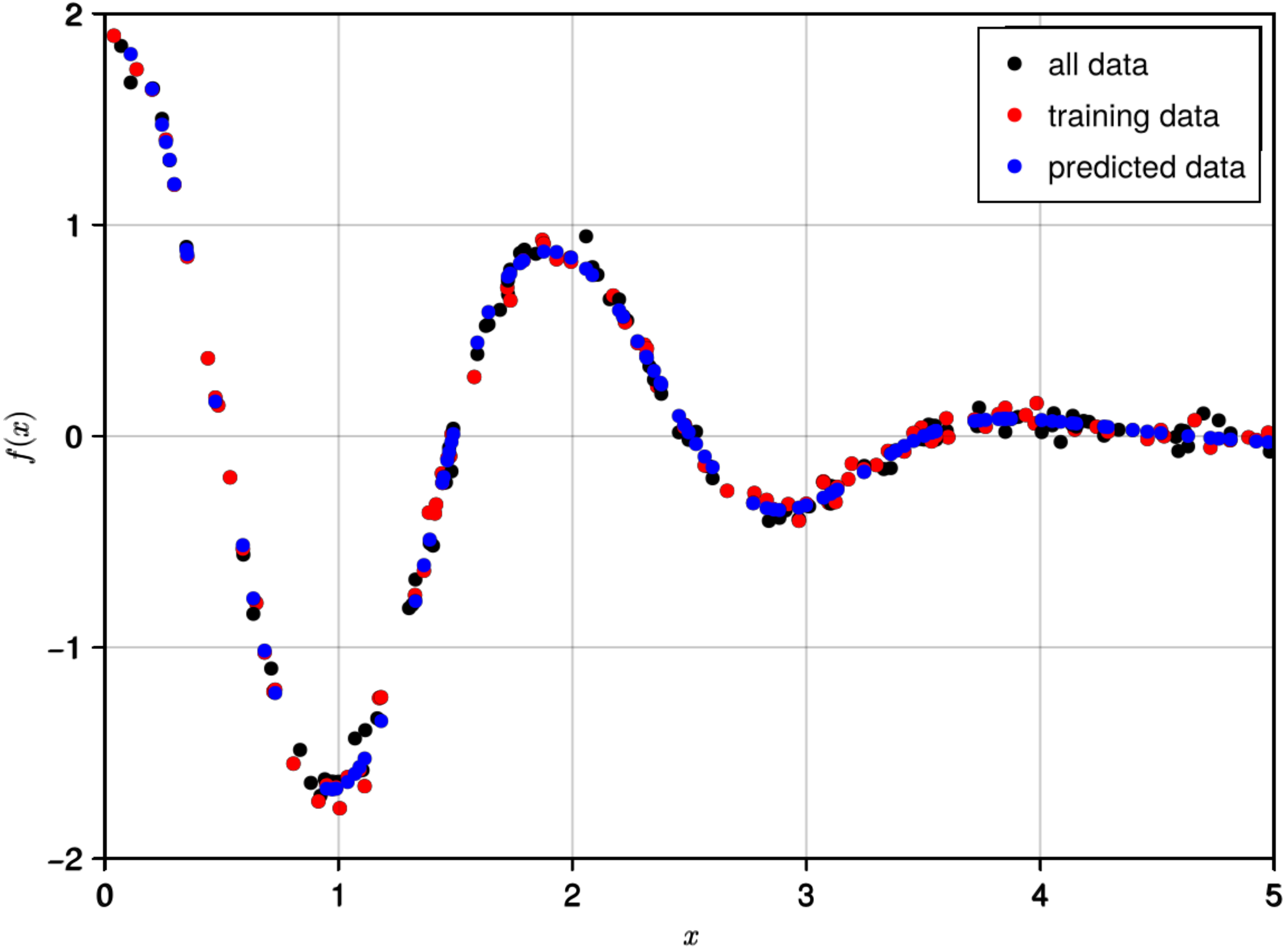


Input Layer $\in \mathbb{R}^1$    Hidden Layer $\in \mathbb{R}^4$    Hidden Layer $\in \mathbb{R}^6$    Output Layer $\in \mathbb{R}^1$

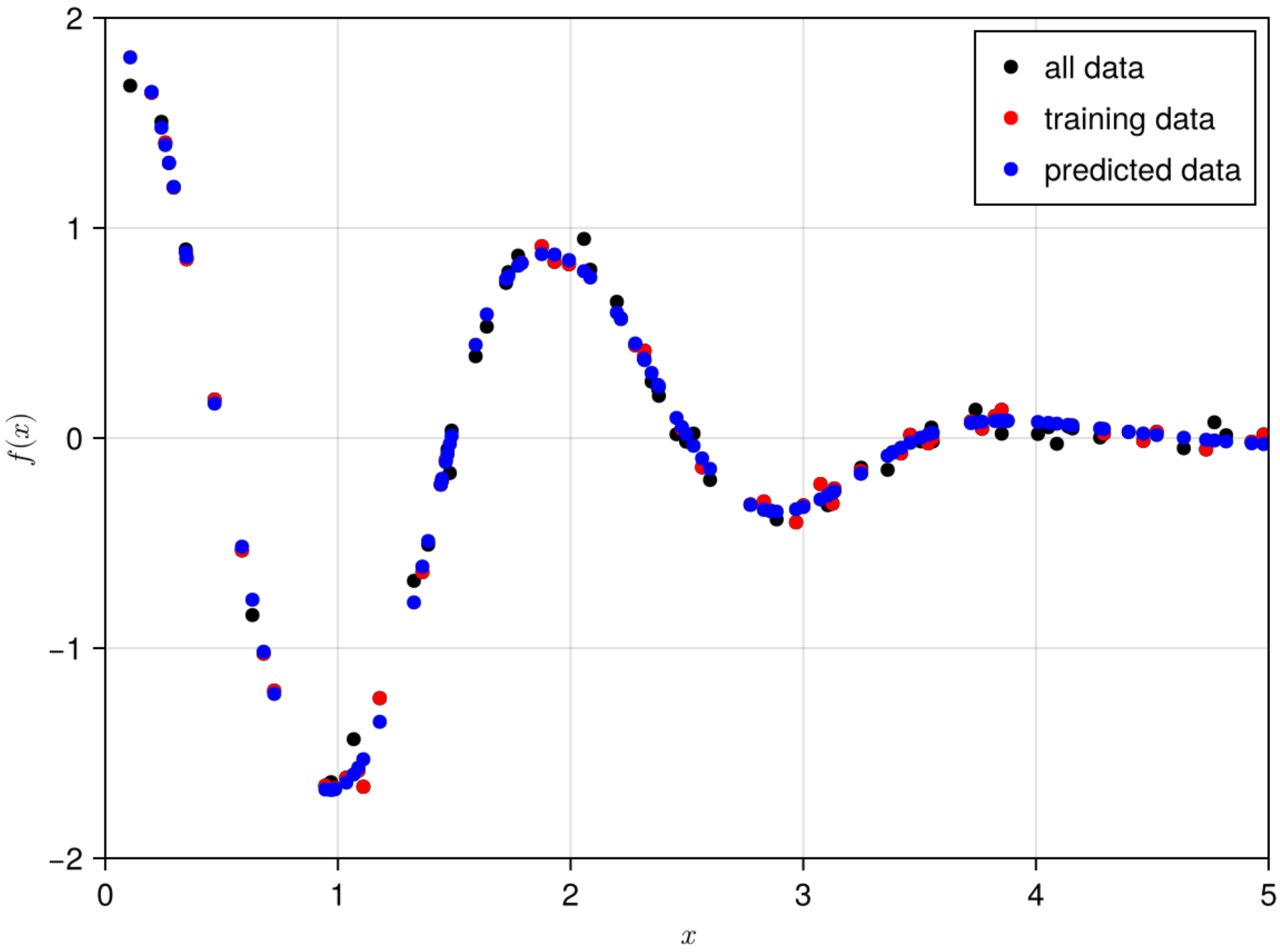
# Training and Validation Region



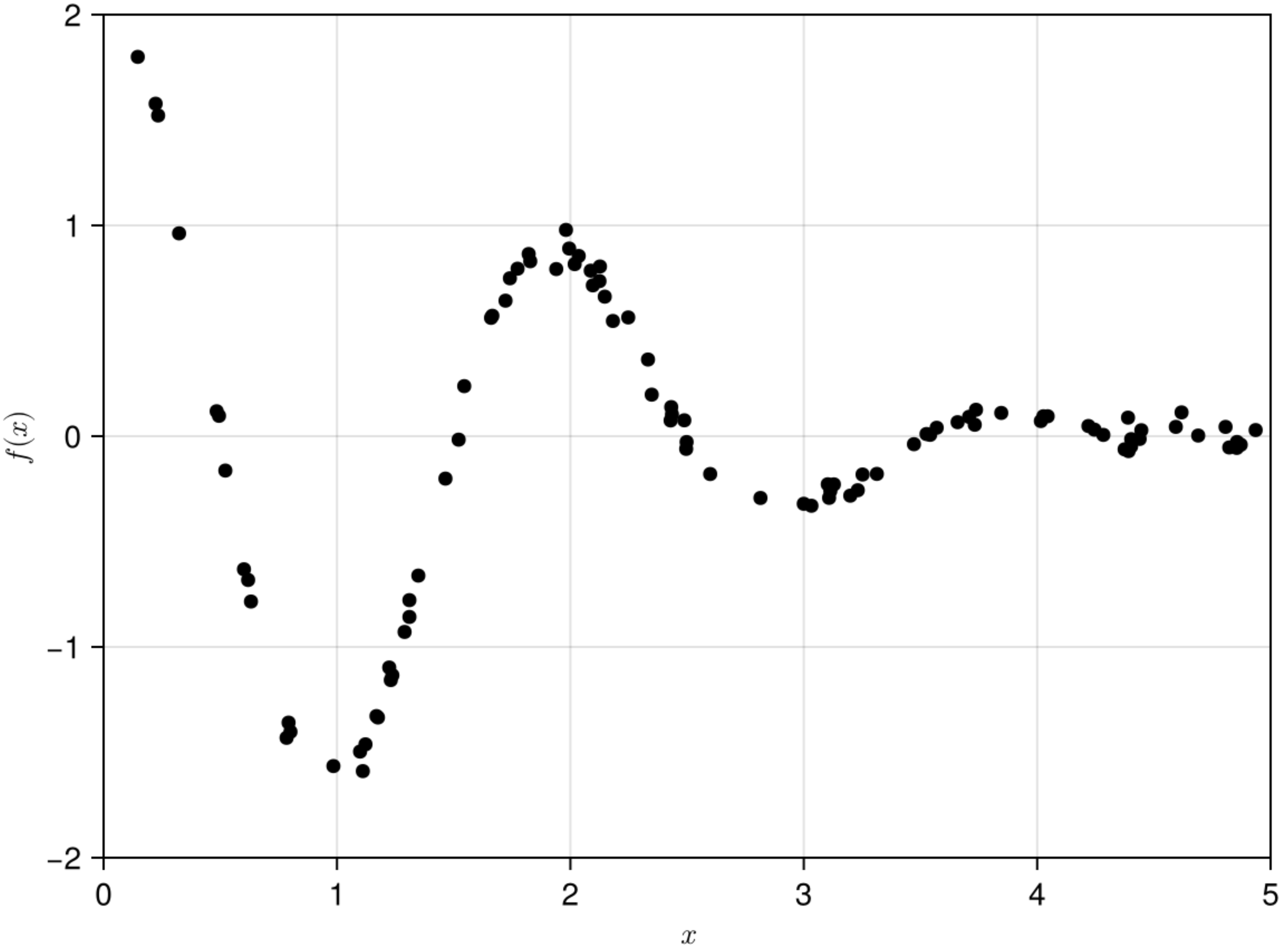
2
1
0
−1
−2
$f(x)$
0
1
2
3
4
5
$x$



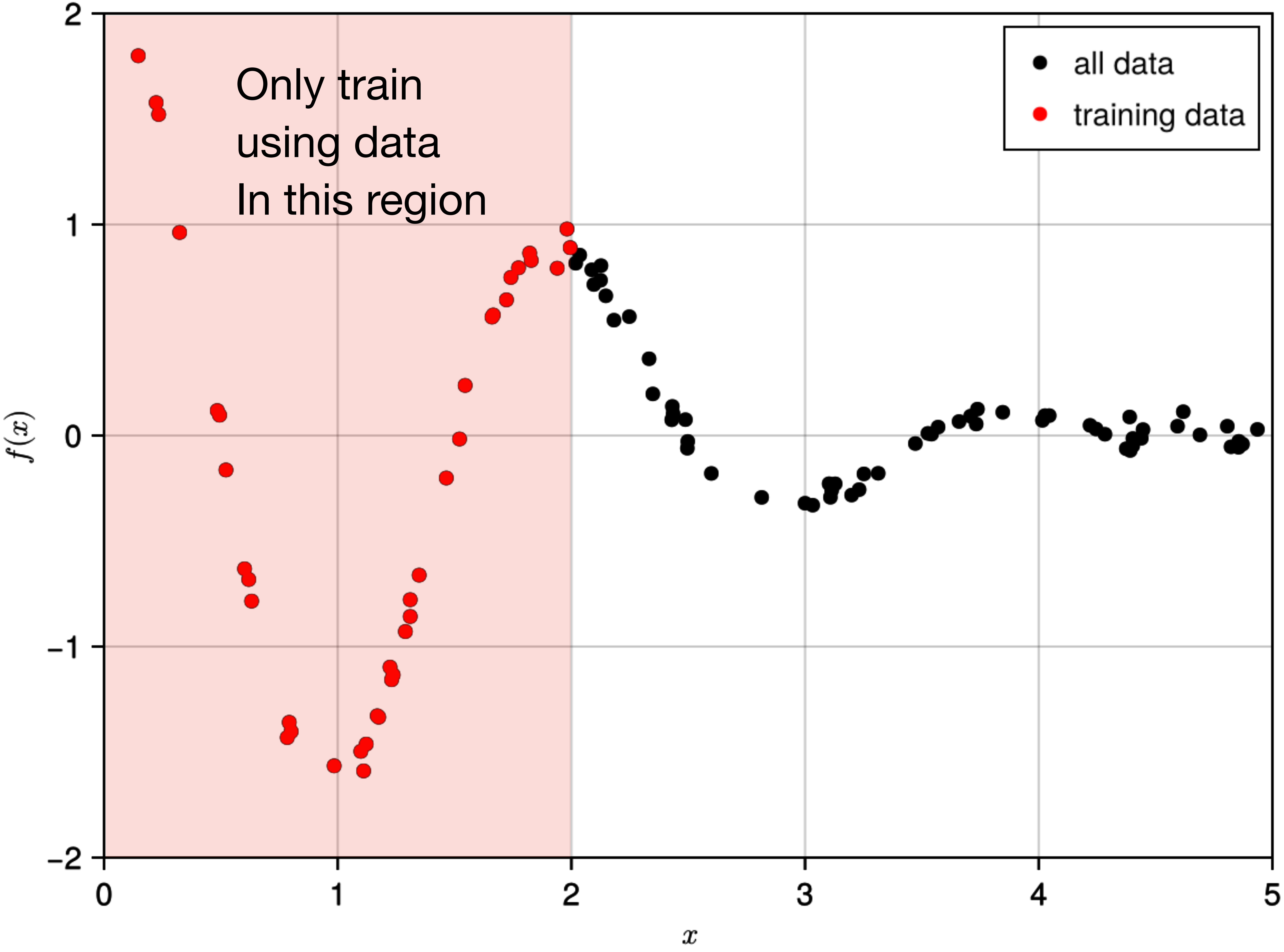
all data
training data
predicted data
2
1
0
−1
−2
$f(x)$
0
1
2
3
4
5
$x$



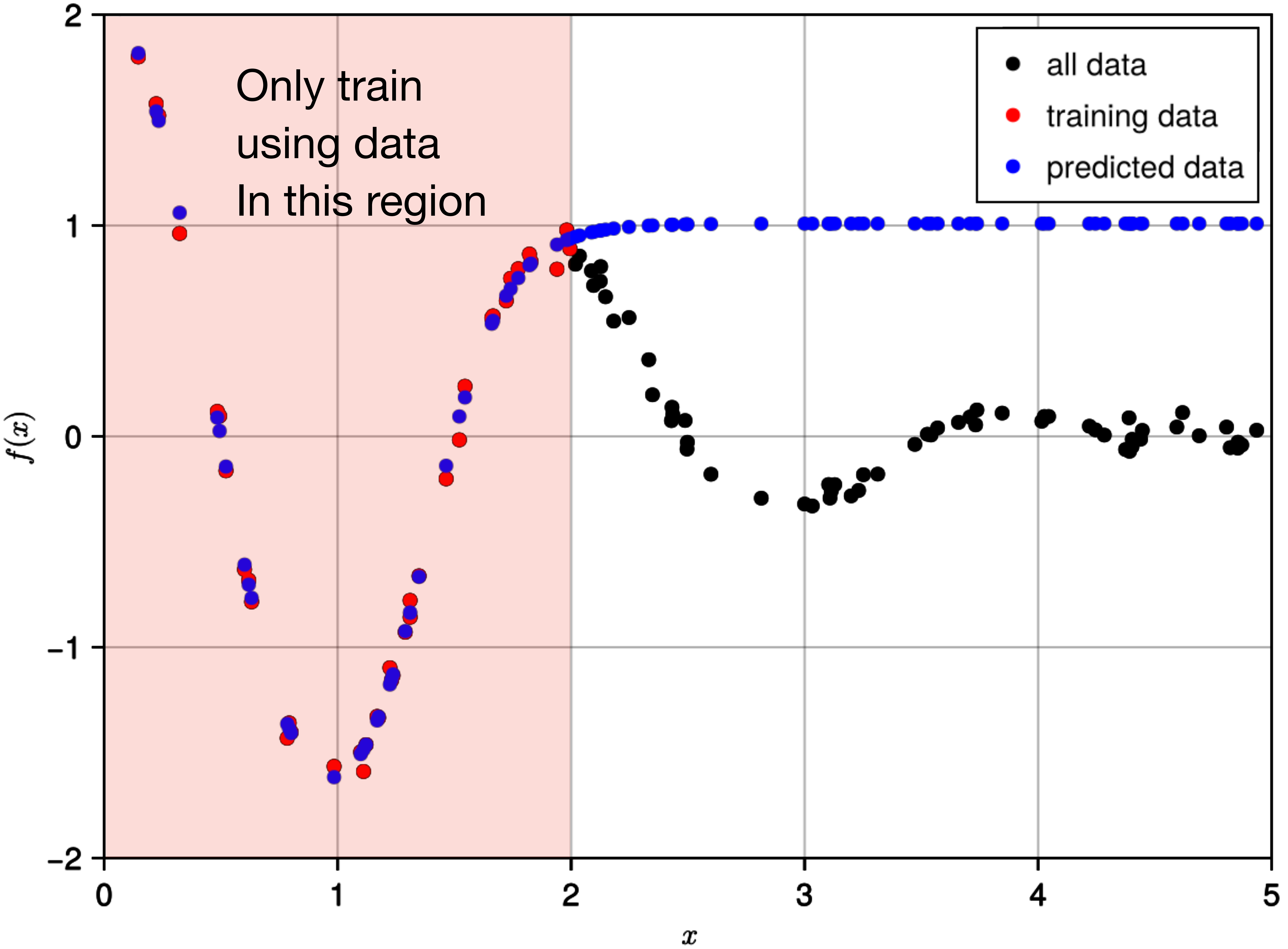
all data
training data
predicted data
$f(x)$
$x$
2
1
0
−1
−2
0
1
2
3
4
5



2
1
0
−1
−2
f(x)
0
1
2
3
4
5
x



Only train
using data
In this region
all data
training data
$f(x)$
$x$
2
1
0
−1
−2
0
1
2
3
4
5



Only train
using data
In this region
all data
training data
predicted data
$f(x)$
$x$
2
1
0
−1
−2
0
1
2
3
4
5